COMP4026 Computer Vision and Pattern Recognition

Written Assignment (2025-26 Second Semester)

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Part (i): Selected YOLO-Family Algorithm — YOLOv7

Selected paper: C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in Proc. IEEE/CVF CVPR, 2023 (arXiv:2207.02696).

Research Problem

The YOLO family (YOLOv4, YOLOv5, YOLOR, YOLOX, PPYOLOE, and Scaled-YOLOv4) by 2022 had made impressive advances in real-time object detection but was still limited by an unsatisfactory Pareto frontier wherein a marginal improvement in MS COCO accuracy usually meant a trade-off among millions more parameters, increased latency, and auxiliary branches which grew the total size of the model. YOLOv7 directly addresses the trilemma of optimizing speed, accuracy, and parameter efficiency in general-purpose object detection with GPUs. The authors state clearly three sub-problems they aim to address. First, common approaches to network scaling—namely depth-only scaling (e.g., EfficientNet) and width-only scaling—are poorly defined for backbones built around concatenation (ELAN / CSPVoVNet), as adding one more computation layer to the bottleneck changes the width of input data passed to the subsequent transition layer. Second, the majority of the recently invented reparameterization techniques (e.g., the "identity plus convolution" branch in RepVGG) are not straightforwardly applicable to backbones having pre-existing shortcut connections (residual connections or concatenations), thus rendering them difficult to apply to YOLO. Third, while auxiliary heads and deep-supervision techniques have been proved effective in the training process, they normally involve adding new inference components or additional labeling tasks to YOLO. Thus, the authors see YOLOv7 as an investigation of optimizations that are trainable only and incur costs solely during the training phase.

Novelty and Key Contributions

Regarding the innovations, I suggest structuring those related to YOLOv7 in terms of four key ideas comprising what is termed a trainable bag of freebies. The first innovation is the Extended Efficient Layer Aggregation Network (E-ELAN). In contrast to ELAN, which performs the aggregation through control over the shortest and longest gradient paths, E-ELAN implements a widening of the cardinality of group convolutions and then a combination of the generated feature maps from all branches in question. With the macro-architecture staying constant as a whole and, therefore, gradient-path structure preserving trainability, the innovation facilitates learning of additional features by the block in question, which is highly important for creating stacks of blocks in deep versions like YOLOv7-E6E.

Secondly, there is compound model scaling specific to concatenation-based architecture scaling. Instead of using separate depth and width scaling approaches, YOLOv7 applies a coupled rule whereby input/output channel ratio in every concatenation block is kept constant throughout the process of scaling. According to ablation analysis conducted for this purpose and discussed in the referenced work, the compound approach provides +1.2 AP boost compared to +0.7 AP increase in case of only width scaling and +1.0 AP increase in case of only depth scaling on COCO dataset.

Thirdly, the paper proposes the idea of planned re-parameterised convolution. Specifically, the authors analyze, using gradient flow theory, the reason for performance degradation in RepConv due to its identity connection placed on top of residual or concatenation connection (ResNet-style or DenseNet-style connection respectively): duplication of existing path decreases gradient diversity. A new version called RepConvN was introduced where such connection is removed, and the authors propose a simple guide stating that whenever input layer receives residual or concatenation connection, it should apply RepConvN.

Finally, the innovation is related to coarse-to-fine label assignment with a lead-head and an auxiliary head. Lead (or final) head creates soft labels which supervise the auxiliary head generating coarse labels (high recall) and itself generating fine labels (high precision). As the auxiliary head does not operate during inference phase, it does not add to computational cost (FLOPs), but it still improves convergence in the process of training. Combined with auxiliary methods such as batch-normalization absorption into convolutions, YOLOR-style implicit knowledge, and exponential moving average (EMA) weight averaging, the model achieves 56.8 AP on MS COCO at over 30 FPS on V100. Thus, YOLOv7 exceeds the transformer-based model SWIN-L Cascade-Mask R-CNN by 509% in terms of speed and by 2 AP, whereas being smaller by approximately 40%. Historically, YOLOv7 is a milestone in the evolution of YOLO since it was the latest "pure vision" YOLO until recently.

Part (ii): Two Unmet Needs of SOTA Visual Object Detection

Modern visual object detectors, including YOLOv7, YOLOv10, Grounding-DINO, YOLO-World, and YOLOE, attain very high accuracy on clean MS COCO datasets and, in some cases, on open vocabulary LVIS. However, after reviewing these approaches in my coursework, I identified two important gaps that remain unsolved by simple enlargement of model sizes.

The first issue is robustness to distribution shifts encountered in the wild. The benchmark of PASCAL-C, COCO-C, and Cityscapes-C has convincingly shown that detectors trained on clean data lose 30%-60% mean average precision when presented with foggy, rainy, snowy, motion-blurred, or darkened input samples. The ICCV 2023 paper COCO-O demonstrates 55.7% relative drop for Faster R-CNN on naturally distributed image inputs. CVPR 2023 extends the issue to robustness of 3D object detectors on KITTI-C, nuScenes-C, and Waymo-C benchmarks, which effectively render camera-only object detectors useless in terms of motion-level corruption. Importantly, neither open vocabulary alignment (such as seen in YOLO-World, Grounding-DINO) nor more powerful backbone network architecture (e.g. ConvNeXt, Swin) helps here since they inherit the weaknesses of ImageNet-style pretrained networks. For application in critical scenarios such as autonomous vehicle perception in nighttime and medical imaging across institutions, this is totally unacceptable. There exists no single detector attaining state-of-the-art accuracy on COCO and robustness to distribution shift on COCO-C and COCO-O benchmarks.

The second problem concerns obtaining state-of-the-art accuracy-efficiency trade-off for inference on edge devices. At the moment, the best-performing detectors in terms of accuracy include transformer-based DINO and Grounding-DINO with more than 200-500 million parameters and without possibility to run real-time on mobile GPU chips like Jetson Orin. YOLO family offers real-time performance at the expense of reduced accuracy. According to recent experiments performed on YOLO26 (arXiv:2509.25164), none of models currently achieves state-of-the-art COCO AP, real-time latency on Jetson-based devices, and compact memory usage. Distillation and quantization techniques partly help to restore accuracy loss, yet the tri-partite dilemma remains open. State-of-the-art detection thus becomes a prerogative of data-center GPUs and not edge devices with cameras onboard.

In my opinion, filling these two important gaps—robust state-of-the-art accuracy and edge device deployability of state-of-the-art detection—is one of the most important frontiers in the area of visual object detection.

References

[1] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2023. arXiv:2207.02696.

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